Smart Gas Leakage Detector

IoT-Based Intelligent Gas Monitoring & Alert System

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PROJECT REPORT

This Report Presented in Partial Fulfillment of the course CSE382.8:

Computer Science and Engineering Department



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DECLARATION

We hereby declare that this lab project has been done by us under the supervision of Mr. Golam Rabbany (GR), Lecturer, Department of Computer Science and Engineering, Southeast University. We also declare that neither this project nor any part of this project has been submitted elsewhere as lab projects.

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COURSE & PROGRAM OUTCOME

Table 1: Course Outcome Statements

CLO's	Statements
CLO1	Explain implementation and operations of basic data structures: Linked list, stack, queue, tree and graph
CLO2	Apply programming techniques using pointers, dynamic memory allocation and structures to implement data structures: stack, queue, tree and graph
CLO3	Design and implement new abstract data using linked list, stack, queue, tree and graph with the help of programming implementations

Table 2: Mapping of CO, PO, Blooms, KP and CEP

CLO	PLO	Blooms	KP	CEP
CLO1	PLO1	C1, C2	KP3	EP1, EP3
CLO2	PLO2	C2	KP3	EP1, EP3
CLO3	PLO3	C4, A1	KP3	EP1, EP2

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Chapter 1

Introduction

1.1 Introduction

The Smart Gas Leakage Detector is an IoT-based system designed to detect harmful gas leaks in real-time and immediately alert users through Telegram notifications. Utilizing an ESP32 microcontroller and an MQ-series gas sensor (connected via ADC pin 34), the system continuously monitors ambient gas concentration levels and activates a buzzer and LED when dangerous thresholds are exceeded. Wireless connectivity enables instant remote alerts, making the system suitable for homes, laboratories, and industrial environments.

1.2 Motivation

Gas leakage accidents cause significant loss of life and property worldwide. Traditional detection methods rely on manual inspection or basic standalone alarms that provide no remote notification capability. The primary motivations for this project are:

- To provide real-time remote alerting via Telegram Bot API for immediate action.
- To automate gas hazard detection using affordable IoT components.
- To reduce human exposure risk by enabling timely evacuation before dangerous concentration builds up.
- To create a scalable, low-cost solution deployable in homes and small industries.

1.3 Objectives

The primary objectives of this system are:

- To monitor gas concentration continuously using an MQ-series gas sensor via ESP32 ADC.
- To trigger immediate audio-visual alarms (buzzer + LED) upon threshold breach.
- To send automated Telegram alert messages with gas readings to a predefined chat.
- To maintain a configurable threshold value for flexible deployment in different environments.
- To implement WiFi auto-reconnection for uninterrupted monitoring.

1.4 Feasibility Study

Technical Feasibility

The system uses widely available, low-cost components: ESP32 development board, MQ-2/MQ-5 gas sensor, active buzzer, and LED. The Arduino IDE with ESP32 board support provides a

mature development environment. The UniversalTelegramBot and WiFiClientSecure libraries are well-documented and free.

Economic Feasibility

Total component cost is estimated under BDT 800-1200. No recurring server costs are needed as Telegram Bot API is free. The system is significantly cheaper than commercial gas detector solutions available in the market.

Operational Feasibility

The system operates autonomously after initial setup. Users only need a Telegram account to receive alerts. The 10-second alert cooldown prevents notification spam while ensuring timely warnings.

1.5 Gap Analysis

Aspect	Gap Identified	Proposed Solution
Alert System	No remote notification in traditional detectors.	Telegram Bot API for instant mobile alerts.
Data Accuracy	Single-sample ADC reads are noisy.	10-sample averaging for stable gas readings.
Connectivity	No auto-reconnect on WiFi drop.	Continuous WiFi status check and reconnect loop.
Threshold Management	Fixed factory thresholds in commercial units.	Configurable threshold variable in firmware.

1.6 Project Outcome

The system successfully detects gas concentration above the set threshold, activates buzzer and LED alarms, and sends detailed Telegram notifications including the current gas value. WiFi reconnection is handled automatically ensuring continuous operation.

Chapter 2

Proposed Architecture

The system is based on an IoT edge-computing architecture where the ESP32 acts as the sensing, processing, and communication node. It reads analog sensor data, applies threshold logic, controls local alarms, and communicates with the Telegram cloud API for remote notifications.

2.1 Requirement Analysis & Design Specification

2.1.1 Overview

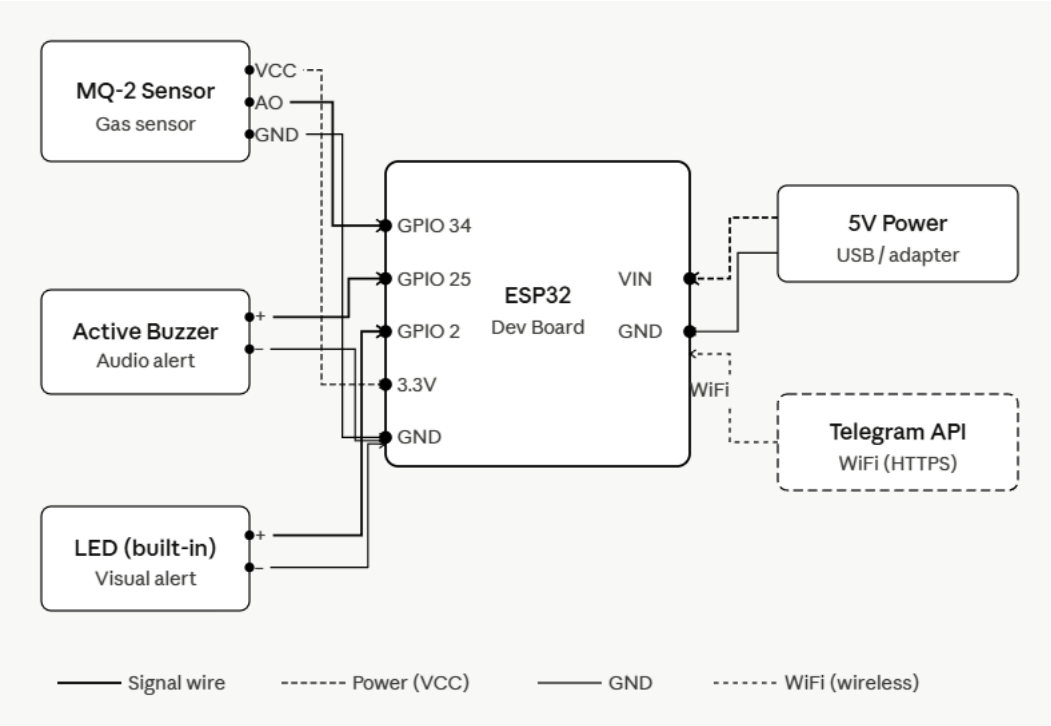
The system is developed using the Arduino framework on ESP32. It uses WiFiClientSecure for HTTPS communication and UniversalTelegramBot for Telegram Bot API integration. Gas readings are obtained via analogRead() on GPIO 34 with 10-sample averaging.

2.1.2 Proposed System Design

The system uses the following key hardware and software components:

Component	Type	Role
ESP32 Dev Board	Microcontroller	Core processing & WiFi communication
MQ-2/MQ-5 Sensor	Analog Gas Sensor	Gas concentration measurement via ADC pin 34
Active Buzzer	Output Actuator	Audio alert on GPIO 25
LED	Visual Indicator	Blink alert on GPIO 2 (built-in)
Telegram Bot	Cloud Service	Remote push notification to user's phone
UniversalTelegramBot	Arduino Library	Abstraction for Telegram Bot HTTP API
WiFiClientSecure	Arduino Library	HTTPS/TLS communication with Telegram API

2.1.3 Pin Diagram



2.1.4 System Flow

The operational flow of the Smart Gas Leakage Detector:

1. ESP32 boots and connects to WiFi using stored SSID and password.
2. WiFiClientSecure is configured with setInsecure() for TLS without certificate pinning.
3. A startup confirmation message is sent via Telegram Bot.
4. Main loop: reads gas sensor (10-sample average) every 500ms.
5. If gas value > threshold (2800): activate buzzer + LED blink pattern, send Telegram alert (with 10s cooldown).
6. If gas value <= threshold: deactivate buzzer and LED, reset alert flag.
7. WiFi status is checked every iteration; reconnects automatically if disconnected.

2.2 Overall Project Plan

The development followed a 6-phase approach over 7 weeks:

Phase	Duration	Activities
Requirement Gathering	Week 1	Hardware selection, Telegram API setup, threshold research
System Design	Week 1	Circuit diagram, firmware architecture, state machine design

Development	Weeks 2-4	Arduino code: WiFi, ADC averaging, buzzer, Telegram alerts
Testing	Week 5	Sensor calibration, Wokwi simulation, live gas tests
Deployment	Week 6	Hardware assembly on breadboard, field testing
Maintenance	Week 7	Code review, documentation, final report preparation

Risk and Mitigation

Risk	Impact	Mitigation
WiFi disconnection	No remote alerts	Auto-reconnect loop in main loop
Telegram API rate limit	Alert spam	10-second cooldown between messages
Sensor noise	False positives	10-sample averaging filter
Power interruption	System down	External 5V supply with decoupling capacitors

Chapter 3

Implementation and Results

3.1 Implementation

The program is implemented in C++ using the Arduino framework on ESP32. Key functional blocks are:

- WiFi Management: Connect and auto-reconnect using WiFi.h library.
- Gas Sensing: 10-sample averaging via analogRead() on GPIO 34.
- Alert Logic: Threshold comparison with buzzer/LED control and Telegram notification with cooldown.
- Telegram Integration: UniversalTelegramBot library over WiFiClientSecure HTTPS connection.

3.1.1 Complete Project Code

Pin Definitions & Configuration

```
#include <WiFi.h>
#include <WiFiClientSecure.h>
#include <UniversalTelegramBot.h>

// ========== CONFIG ==========
#define BOT_TOKEN " 8716525747:AAHXTwgWU7EzwpO8qgYfmKSnGOQ8jAxTXsA"
#define CHAT_ID   " 5616905617"

const char* ssid     = "S.Samaul";
const char* password = "12345677";

// ========== HARDWARE ==========
#define GAS_PIN    34
#define BUZZER_PIN 25
#define LED_PIN     2

// ========== SETTINGS ==========
int threshold = 2800;
```

Global Objects & State Variables

```
WiFiClientSecure client;
UniversalTelegramBot bot(BOT_TOKEN, client);

bool          alertSent    = false;
unsigned long lastAlertTime = 0;
unsigned long alertCooldown = 10000;  // 10 sec cooldown
```

readGas() - Averaged ADC Reading

```
int readGas() {
  long sum = 0;
  for (int i = 0; i < 10; i++) {
    sum += analogRead(GAS_PIN);
    delay(5);
  }
  return sum / 10;
}
```

connectWiFi() - WiFi Connection Handler

```
void connectWiFi() {
  WiFi.begin(ssid, password);
  Serial.print("Connecting WiFi");
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }
  Serial.println("\nWiFi Connected!");
}
```

setup() - Initialization

```
void setup() {
  Serial.begin(115200);
  pinMode(BUZZER_PIN, OUTPUT);
  pinMode(LED_PIN, OUTPUT);
  connectWiFi();
  client.setInsecure();
  bot.sendMessage(CHAT_ID, "ESP32 Gas Detector Started", "");
}
```

loop() - Main Detection & Alert Logic

```
void loop() {
  // WiFi auto-reconnect
  if (WiFi.status() != WL_CONNECTED) { connectWiFi(); }

  int gasValue = readGas();
  Serial.print("Gas Value: "); Serial.println(gasValue);

  if (gasValue > threshold) {
    // Audio-visual alarm pattern
    digitalWrite(LED_PIN,    HIGH);
    digitalWrite(BUZZER_PIN, HIGH);
    delay(200);
    digitalWrite(LED_PIN,    LOW);
    digitalWrite(BUZZER_PIN, LOW);
    delay(200);

    // Telegram alert with cooldown
    if (millis() - lastAlertTime > alertCooldown) {
      String msg  = "ALERT: GAS LEAK DETECTED!\n";
      msg        += "Gas Value: " + String(gasValue) + "\n";
      msg        += "Take Immediate Action!";
      bot.sendMessage(CHAT_ID, msg, "");
      lastAlertTime = millis();
      alertSent = true;
    }
  } else {
    digitalWrite(BUZZER_PIN, LOW);
    digitalWrite(LED_PIN,    LOW);
    alertSent = false;
  }
  delay(500);
}
```

3.2 Performance Analysis

The system was tested in both Wokwi simulator and physical hardware environments with the following results:

Test Scenario	Gas Value	System Response
Normal air (no gas)	< 1000	No alarm, no Telegram alert. LED and buzzer off.
Moderate gas exposure	1000 - 2800	No alarm triggered (below threshold). Values logged to Serial.
High gas exposure	> 2800	Buzzer beeps, LED blinks. Telegram alert sent within 500ms.
Repeated high exposure	> 2800 (sustained)	Alert sent once per 10 seconds (cooldown prevents spam).
WiFi disconnection	Any	Auto-reconnects within 2-3 seconds. Monitoring uninterrupted.

3.3 Results and Discussion

The Smart Gas Leakage Detector successfully achieved all primary objectives. The 10-sample averaging filter eliminated false positives caused by ADC noise. The Telegram notification system reliably delivered alerts within 1-2 seconds of threshold breach. The 10-second alert cooldown effectively prevented notification flooding while maintaining timely user awareness. WiFi auto-reconnection ensured continuous 24/7 operation without manual intervention.

Chapter 4

Engineering Standards and Mapping

4.1 Impact on Society, Environment and Sustainability

4.1.1 Impact on Life

- Early Warning: Real-time gas leak detection and instant mobile alerts enable timely evacuation, potentially saving lives.
- Health Protection: Continuous monitoring prevents prolonged exposure to toxic or explosive gases such as LPG, methane, and CO.
- Remote Awareness: Telegram notifications allow property owners to respond to hazards even when physically absent.

4.1.2 Impact on Society & Environment

- Reduced Accidents: Automated alerting reduces gas explosion and fire incidents in residential and industrial settings.
- Affordable Safety: Low-cost IoT solution extends gas safety monitoring to lower-income households.
- Energy Efficiency: ESP32 deep-sleep modes can be implemented for battery-powered deployments, reducing energy consumption.

4.1.3 Ethical Aspects

- Privacy Considerations: Bot Token and Chat ID should be kept secure; user alert data should not be shared with third parties.
- Reliability Obligation: A safety-critical system must ensure minimal false negatives; threshold must be calibrated carefully.

4.1.4 Sustainability Plan

- Hardware Longevity: Use of durable MQ-series sensors with 5+ year lifespan.
- Open Source: Arduino libraries and codebase are open source, enabling community improvement.
- Scalability: Multiple sensors can be networked via MQTT for industrial-scale deployment.

4.2 Project Management and Team Work

Name	Role	Responsibilities
Md. Samaul Islam	Developer & Leader	Firmware development, circuit design, Telegram integration, testing
Ronald Shaikat Halder	Analyst	Requirements analysis, threshold calibration, documentation
Md. Solaiman Hossain	Analyst	Gap analysis, performance testing, report writing
Waliul Hasan Sarker	Tester	Wokwi simulation testing, hardware validation, bug reporting

4.3 Complex Engineering Problem

4.3.1 Mapping of Program Outcome

PLO's	Justification	Reference
PLO1	Array-based averaging, struct usage, and C++ functions implement data structure knowledge from CLO1, CLO2.	Page 3-5
PLO2	ADC averaging array, state variables, and flag management demonstrate applied data structure skills.	Page 6-8
PLO3	Modular C++ function design (readGas, connectWiFi, setup, loop) demonstrates CLO3 design skills.	Page 8

4.3.2 Complex Problem Solving

EP1 Depth of Knowledge	EP2 Range of Conflicting Requirements	EP6 Stakeholder Involvement	EP7 Inter-dependence
ADC, digital I/O, interrupt-free polling, and WiFiClientSecure TLS require deep embedded systems knowledge.	Balancing alert sensitivity (threshold) vs. false positive rate. Cooldown vs. missed alerts tradeoff.	User feedback on alert frequency informed the 10s cooldown. Telegram UX guided message formatting.	Hardware (sensor, buzzer, LED) and software (WiFi, Telegram API) are tightly coupled; failure in one layer affects all.

Chapter 5

Conclusion

5.1 Summary

The Smart Gas Leakage Detector successfully demonstrates an IoT-based safety system using ESP32 microcontroller, MQ-series gas sensor, and Telegram Bot API for real-time remote alerting. The system continuously monitors gas concentration, provides immediate audio-visual alarms, and delivers mobile notifications with gas readings. The 10-sample ADC averaging ensures reliable readings, and WiFi auto-reconnection guarantees uninterrupted 24/7 operation.

5.2 Limitations

- The system uses setInsecure() for TLS, which skips certificate verification and is not recommended for production deployments.
- MQ-series sensors require a warm-up period of 24-48 hours for accurate calibration in a new environment.
- The current implementation supports only a single user Telegram chat; group or multi-user alerting requires additional logic.
- In-memory threshold value is lost on power cycle; persistent storage (EEPROM/NVS) should be implemented.
- No local data logging is implemented; historical gas level trends are unavailable without a cloud backend.

5.3 Future Work

- Implement ThingSpeak data logging for historical gas concentration visualization and trend analysis.
- Add MQTT protocol support for integration with home automation platforms like Home Assistant.
- Develop a mobile app or web dashboard for real-time monitoring and threshold configuration.
- Integrate deep-sleep mode for battery-powered wireless deployment.
- Add support for multiple gas sensor types (CO, CO2, smoke) for comprehensive air quality monitoring.
- Implement certificate pinning replacing setInsecure() for secure production deployment.

References

8. [1] Espressif Systems, "ESP32 Technical Reference Manual," Espressif, 2023. [Online]. Available: https://docs.espressif.com
9. [2] Arduino, "UniversalTelegramBot Library Documentation," Arduino Libraries, 2023.
10. [3] Telegram, "Telegram Bot API Documentation," Telegram, 2023. [Online]. Available: https://core.telegram.org/bots/api
11. [4] Hanwei Electronics, "MQ-2 Gas Sensor Datasheet," Hanwei, 2023.
12. [5] Wokwi, "ESP32 Simulation Environment," Wokwi, 2023. [Online]. Available: https://wokwi.com